

MASA Hackathon 2026

R-Ignite

Plots, Tables & Analytical Findings
Climate Risk Assessment for a Multinational Reinsurer

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Document overview

This document is a visual and tabular companion to the main 10-page report. It consolidates all 13 figures and the key analytical tables produced in the course of the analysis, organised by analytical theme. Each figure is accompanied by a one-line caption explaining what to look at and what conclusion the team draws from it. Each table is preceded by a brief context note. Numbers in this document are reproducible by running the Python pipeline documented in *code_documentation.md*.

Structure

- • 1. Project context and indicator panel
- • 2. SEA emissions trajectory and country profiles
- • 3. Driver analysis - correlations, partial correlations, sign flips
- • 4. Country idiosyncratic risk - STIRPAT residuals
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1. Project context and indicator panel

The MASA Hackathon 2026 brief asks teams to assess climate-related risks for a multinational reinsurance client by using the World Bank's World Development Indicators (WDI) to identify drivers, forecast 2024 emissions, link emissions to insurance claims, and stress-test a 2030 mitigation strategy. We frame the engagement around three TCFD-aligned risk channels - **transition**, **physical**, and **adaptive capacity** - and curate a panel of 16 indicators from the WDI's 1,486-indicator universe.

Table 1.1 - Curated indicator panel (n = 16)

#	Indicator (short name)	WDI code	Category
1	GHG_total_MtCO2e (TARGET)	EN_GHG_ALL_MT_CE_AR5	Emissions
2	GHG_per_capita_tCO2e	EN_GHG_ALL_PC_CE_AR5	Emissions
3	CO2_per_capita_tCO2e	EN_GHG_CO2_PC_CE_AR5	Emissions
4	CO2_intensity_GDP	EN_GHG_CO2_RT_GDP_KD	Emissions
5	renewable_energy_pct	EG_FEC_RNEW_ZS	Energy
6	renewable_elec_pct	EG_ELC_RNEW_ZS	Energy
7	energy_use_pc	EG_USE_PCAP_KG_OE	Energy
8	forest_area_pct	AG_LND_FRST_ZS	Land Use
9	agri_land_pct	AG_LND_AGR_I_ZS	Land Use
10	GDP_constant_2015USD	NY_GDP_MKTP_KD	Macro
11	GDP_per_capita_2015USD	NY_GDP_PCAP_KD	Macro
12	population	SP_POP_TOTL	Macro
13	urban_pop_pct	SP_URB_TOTL_IN_ZS	Macro
14	industry_pct_GDP	NV_IND_TOTL_ZS	Sectoral
15	agriculture_pct_GDP	NV_AGR_TOTL_ZS	Sectoral
16	freshwater_withdrawal_pct	ER_H2O_FWTL_ZS	Water

Table 1.2 - SEA scope: 10 economies, all with full coverage

Country	Population (~m)	GDP (~USD bn, 2024)	2024 GHG (Mt CO2e)
Indonesia	280	1,400	1,324
Vietnam	100	430	584
Thailand	70	530	422
Malaysia	34	430	332
Philippines	115	470	267
Myanmar	54	67	118
Singapore	5.9	500	76
Cambodia	17	47	50
Lao PDR	7.6	17	42
Brunei Darussalam	0.46	17	12

2. SEA emissions trajectory and country profiles

The headline pattern in the regional data is **Vietnam's rapid emissions growth**, which has lifted it past Thailand to become SEA's second-largest emitter. From 152 Mt CO₂e in 2000 to 584 Mt CO₂e in 2024 (3.8x growth), Vietnam now sits second only to Indonesia. Philippines, by contrast, grew 1.7x over the same period - a divergence that anchors the country-comparison case study.

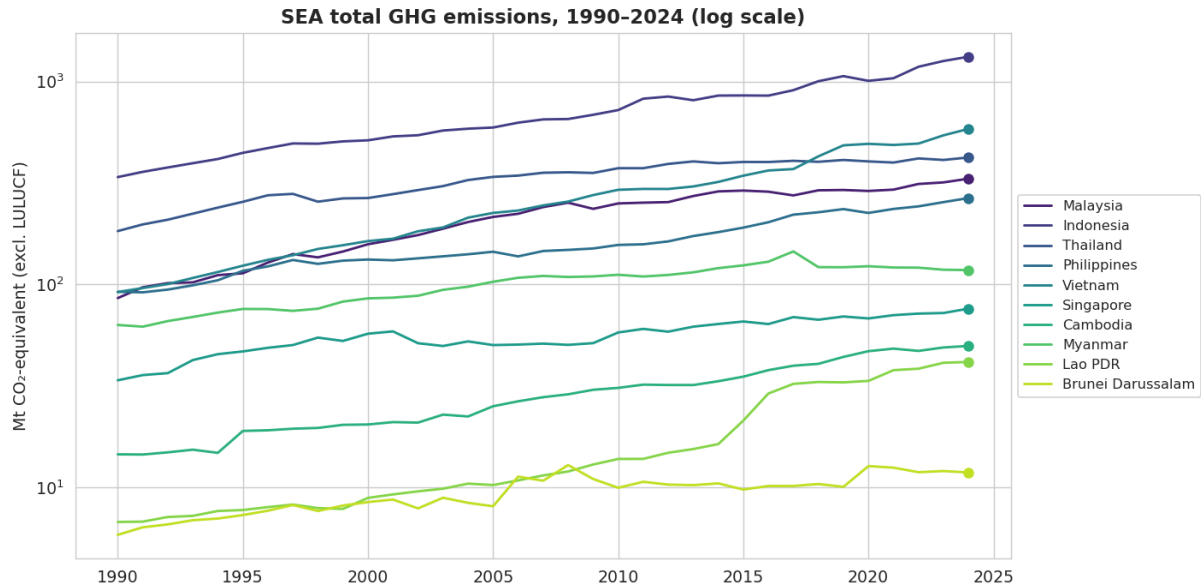


Figure 1. Total GHG emissions in SEA on a log scale, 1990-2024. Indonesia leads the region by a wide margin; Vietnam overtook Thailand in 2018.

3. Driver analysis - correlations and partial correlations

Before building forecast models, we identify which indicators carry information about emissions *after* controlling for the dominant scale drivers (GDP and population). The pairwise correlation answers '*how does this indicator move with GHG?*'; the partial correlation answers '*how does this indicator move with GHG after removing scale?*'

Table 3.1 - Pairwise correlations with log(GHG), SEA panel 2000-2023

Indicator	Pairwise r	p-value
GDP_constant_2015USD	+0.772	<0.001
population	+0.758	<0.001
agri_land_pct	+0.673	<0.001
forest_area_pct	-0.493	<0.001
energy_use_pc	-0.452	<0.001
CO2_intensity_GDP	+0.413	<0.001
GDP_per_capita_2015USD	-0.367	<0.001
renewable_elec_pct	-0.143	0.028
industry_pct_GDP	-0.137	0.034
renewable_energy_pct	-0.133	0.040
urban_pop_pct	-0.069	0.284
freshwater_withdrawal_pct	-0.061	0.357
agriculture_pct_GDP	-0.024	0.708

Table 3.2 - Partial correlations (after controlling for log GDP and log population)

Indicator	Partial r	Change vs pairwise
CO2_intensity_GDP	+0.754	Strengthens (was +0.413)
forest_area_pct	+0.634	SIGN FLIP (was -0.493)
industry_pct_GDP	+0.539	SIGN FLIP (was -0.137)
energy_use_pc	+0.517	SIGN FLIP (was -0.452)
renewable_energy_pct	-0.437	Strengthens (was -0.133)
GDP_per_capita_2015USD	-0.433	Strengthens (was -0.367)
freshwater_withdrawal_pct	-0.362	Strengthens (was -0.061)
renewable_elec_pct	-0.239	Strengthens (was -0.143)
urban_pop_pct	-0.228	Strengthens (was -0.069)
agriculture_pct_GDP	+0.161	Sign flip (was -0.024)
agri_land_pct	+0.063	Collapses (was +0.673)

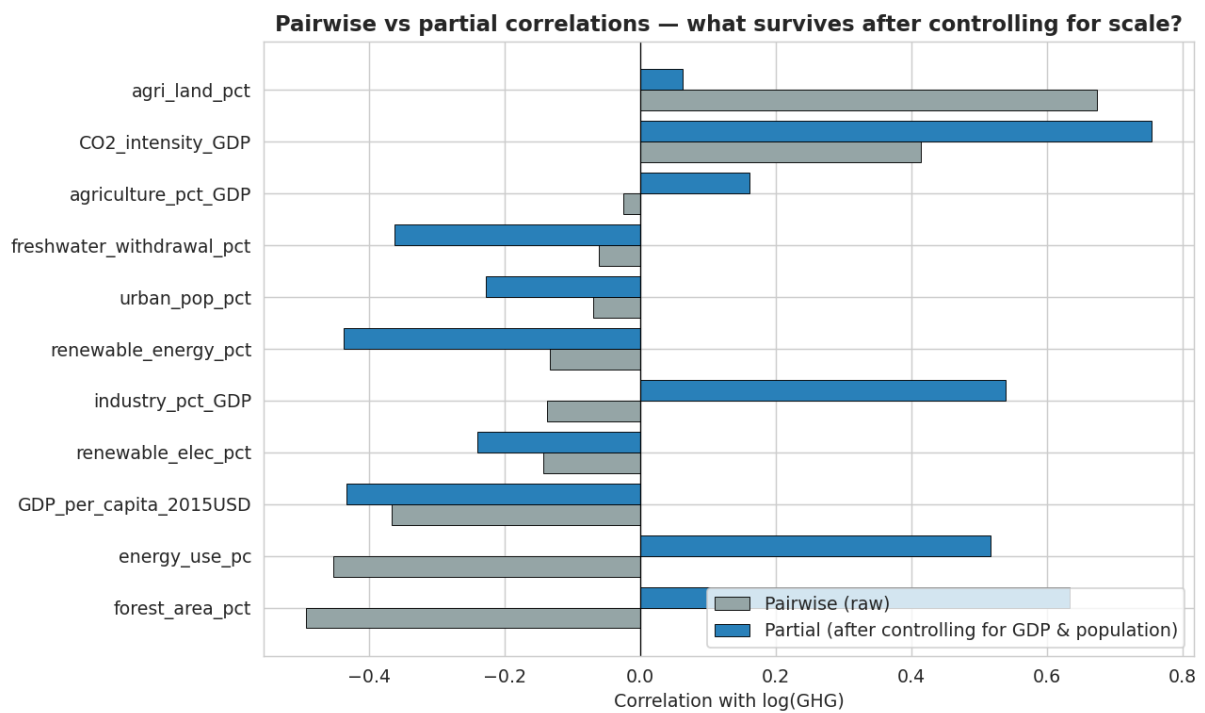


Figure 2 (was fig9). Pairwise (grey) vs partial (blue) correlations with log(GHG). Three sign-flips are visible: forest area, industry share, and energy use per capita all reverse direction once scale is controlled.

Key insight. Three of our 16 indicators flip sign once scale is controlled. Forest-rich SEA economies emit MORE than predicted - a deforestation-and-land-use channel hidden by the LULUCF-excluding indicator. This is the analytically distinctive finding most teams will miss.

4. Country idiosyncratic risk - STIRPAT residuals

The STIRPAT framework (Dietz & Rosa 1997) regresses log(emissions) on log(population) and log(GDP). Our global regression on 1990-2023 data fits with $R^2 = 0.937$, with population coefficient 0.580 and GDP coefficient 0.539. The **residuals** from this regression for SEA economies (averaged 2019-2023) identify which countries emit more or less than their scale would predict - the strongest available proxy for idiosyncratic transition-risk exposure.

Table 4.1 - STIRPAT residuals by country (SEA, 2019-2023 average)

Country	Residual (%)	Interpretation
Brunei Darussalam	+287.1	Severe over-emitter (LNG-dominated)
Lao PDR	+92.9	Strong over-emitter
Malaysia	+32.2	Moderate over-emitter
Vietnam	+24.4	Moderate over-emitter (key portfolio risk)
Cambodia	+13.2	Mild over-emitter
Thailand	+5.6	Near scale prediction
Myanmar	+4.1	Near scale prediction
Singapore	-11.5	Mild under-emitter
Indonesia	-18.8	Moderate under-emitter
Philippines	-49.2	STRONG under-emitter (favourable profile)

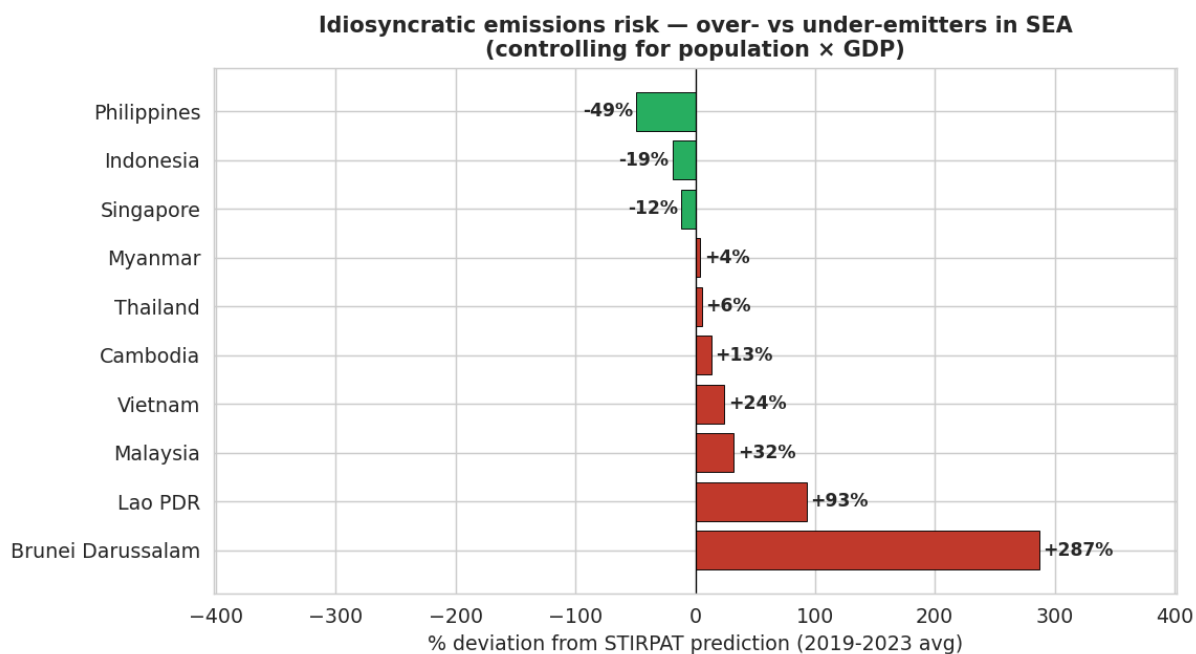


Figure 3 (was fig10). SEA STIRPAT residuals 2019-2023. Red bars = over-emitters (emit more than population × GDP would predict); green = under-emitters. The Philippines vs Vietnam contrast (-49 % vs +24 %) anchors the country-comparison case study.

5. Sectoral residual decomposition

Aggregate country residuals are useful for tiering but conceal the sector that is driving each country's deviation. We re-run STIRPAT on eight sectoral GHG aggregates (Power Industry, Industrial Combustion, Industrial Processes, Transport, Agriculture, Waste, Buildings, Fugitive Emissions). The result decisively reshapes the Vietnam story: the +24 % aggregate over-emission concentrates almost entirely in Power Industry (+280 %) and Industrial Combustion (+276 %). Vietnam's transition risk is therefore **a power-sector story**, which directly informs the parametric product design in the recommendations section.

Table 5.1 - Sectoral STIRPAT residuals (% deviation from prediction, 2019-2023 avg)

Country	Power Ind	Ind Comb	Ind Proc	Transport	Agri	Waste	Buildings	Fugitive
Malaysia	+172	+78	+51	+62	-54	+16	-55	+154
Indonesia	+77	+65	-21	+17	-39	-6	-67	+83
Thailand	+63	+90	+155	+66	+18	+50	-44	-16
Philippines	+66	-54	-36	-22	-37	-46	-61	-67
Vietnam	+280	+276	+147	+8	+26	-9	-43	-24
Singapore	-46	+9	+255	-76	-99	+24	-88	+17
Cambodia	+38	+29	+122	+70	+146	-40	-37	-69
Myanmar	+14	+34	-19	+4	+110	-38	-1	-76
Lao PDR	+781	+17	+420	+18	+189	-50	-60	+14
Brunei Darussalam	+351	+76	+137	+25	-55	-42	-51	+4,205

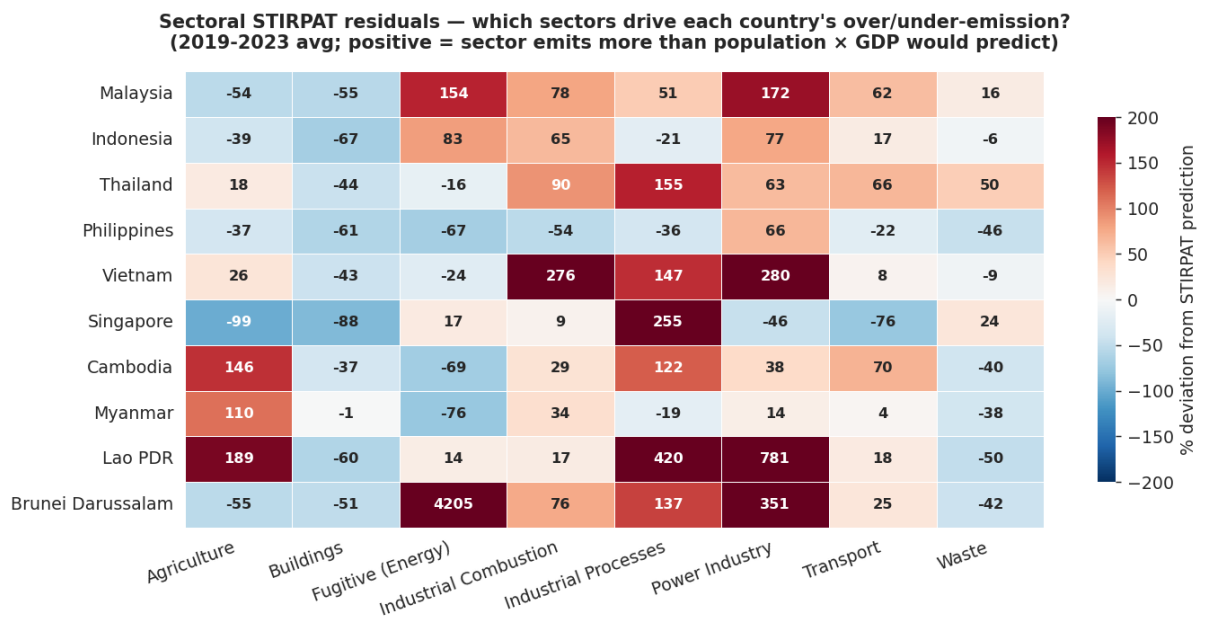


Figure 4 (was fig12). Sectoral STIRPAT residuals heatmap. Red = sector emits more than scale predicts; blue = less. Vietnam (Power Industry +280 %, Industrial Combustion +276 %) and Lao PDR (Power Industry +781 %) are the most concentrated transition-risk hotspots.

6. Decoupling diagnostic

The clearest socio-economic indicator for adaptive capacity is whether GDP-per-capita growth is decoupling from per-capita emissions. The trailing 5-year cross-country correlation between $\log(\text{GDP/cap})$ and CO_2/cap in SEA fell from $r = 0.92$ in 2005 to $r = 0.81$ in 2023 - directionally encouraging but well short of the decoupling threshold ($r \leq 0.3$) seen in mature OECD economies. Only Singapore and Malaysia show country-specific decoupling on a per-capita basis; the remaining eight economies remain firmly coupled.

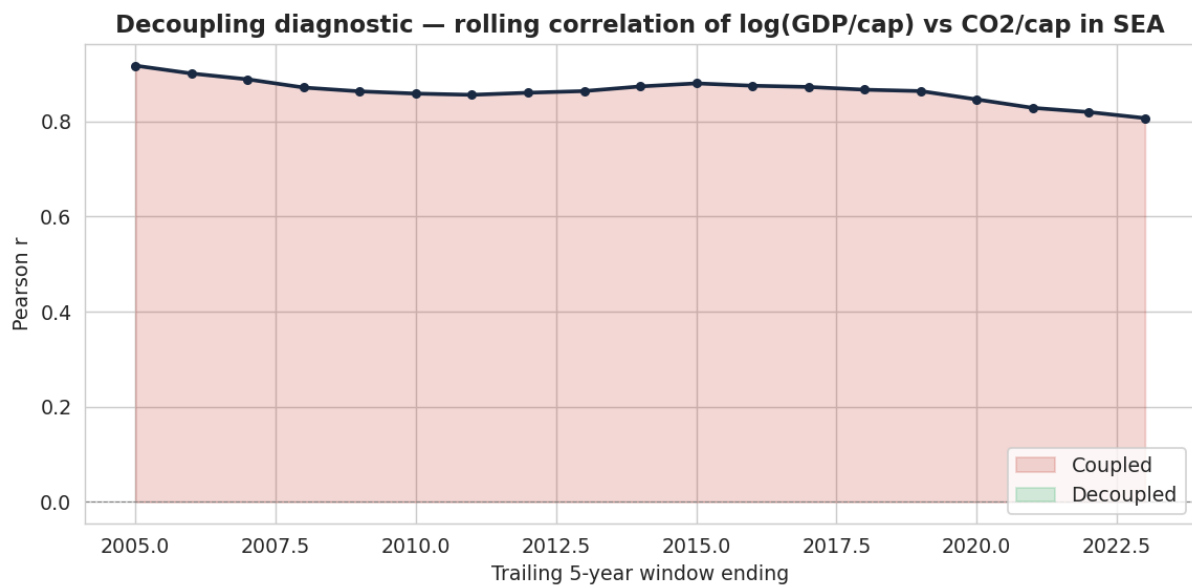


Figure 5 (was fig11). Trailing 5-year correlation of $\log(\text{GDP/cap})$ vs CO_2/cap in SEA. The slow downward drift is real but well above the decoupling threshold.

Implication. The scale effect will continue to dominate SEA emissions through 2030 - supporting our forecast structure (autoregressive XGBoost on global panel). Mitigation cannot be assumed; it has to be policy-driven (NGFS Net Zero) or product-driven (the proposed renewable-linked reinsurance lever).

7. Forecasting models - M1, M2, M3 comparison

We benchmark four candidate model specifications on the 2024 hold-out, selecting the autoregressive XGBoost variant for production forecasting and the structural-only XGBoost for driver interpretation:

Table 7.1 - Out-of-sample 2024 MAPE comparison (SEA region)

Model	SEA 2024 MAPE	Use case
M1: Per-country log-linear	9.23 %	Sanity-check baseline
M2: Auto-ARIMA	2.67 %	Transparent companion
M3a: XGBoost (autoregressive)	2.18 %	SELECTED forecast model
M3b: XGBoost (structural-only)	6.73 %	Driver interpretation

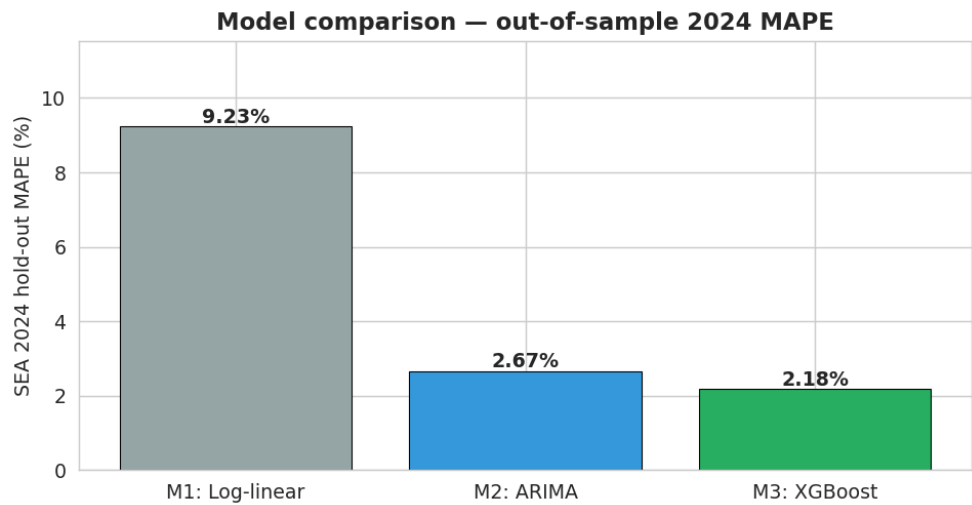


Figure 6 (was fig2). MAPE comparison across the three benchmarked specifications.

Table 7.2 - XGBoost feature importance (autoregressive variant)

Feature	Gain
log_GHG_lag1	0.613
log_GHG_lag2	0.347
log_GDP	0.023
log_pop	0.015
CO2_intensity_GDP	0.001
urban_pop_pct	0.000
Other features	<0.001

Table 7.3 - XGBoost feature importance (structural-only variant)

Feature	Gain
log_pop	0.543
log_GDP	0.307
industry_pct_GDP	0.036
CO2_intensity_GDP	0.030
energy_use_pc	0.023
forest_area_pct	0.021
renewable_energy_pct	0.013
GDP_per_capita_2015USD	0.013
urban_pop_pct	0.010
agriculture_pct_GDP	0.003

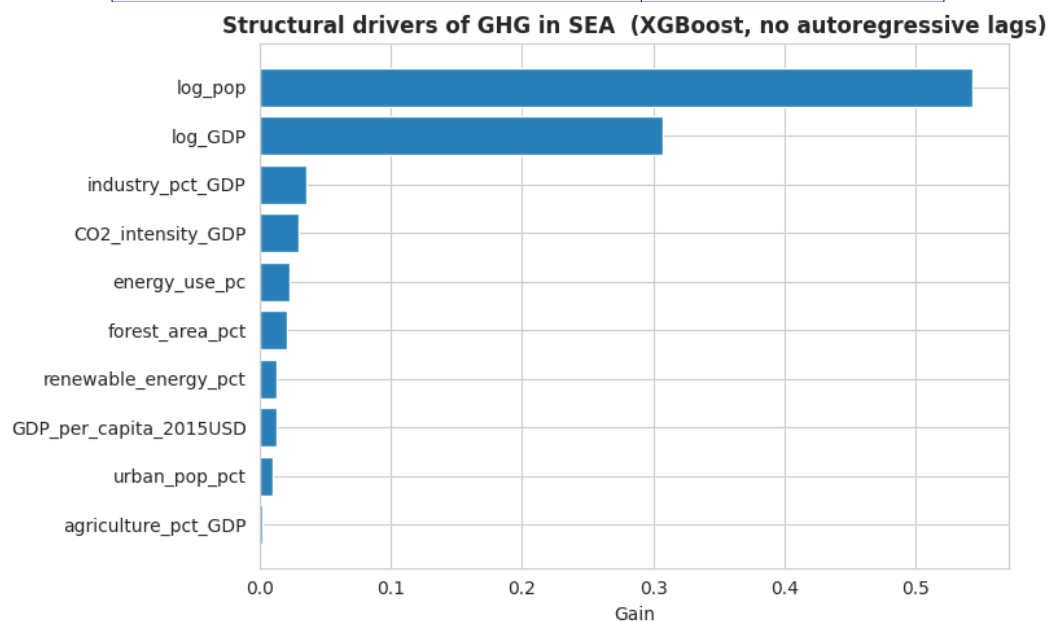


Figure 7 (was fig8). Structural feature importance. Population and GDP dominate the driver story (combined 0.85), validating the STIRPAT framework.

8. Two-way fixed-effects robustness check

Partial correlations and STIRPAT residuals identify drivers but treat observations as independent. We re-estimate the structural model under three specifications - pooled OLS, country FE only, and two-way FE (country + year) - with standardised covariates and country-cluster-robust standard errors. **Three drivers survive the most demanding spec:** log(GDP), CO2 intensity of GDP, and urban population share. Renewable energy share is revealed as a cross-sectional signal only.

Table 8.1 - Standardised coefficients across three specifications (SEA panel 1995-2023, n = 288)

Indicator (standardised)	Pooled OLS	Country FE	Two-way FE	Robust?
log_GDP	0.776***	0.597***	0.608**	YES
log_pop	0.692***	0.637 ns	0.500 ns	Cross-section only
CO2_intensity_GDP	0.244***	0.161***	0.167***	YES
urban_pop_pct	0.128**	0.248***	0.241**	YES (strengthens)
forest_area_pct	0.053**	0.146**	0.124 ns	Borderline
renewable_energy_pct	0.172*	-0.034 ns	-0.038 ns	Cross-section only
industry_pct_GDP	0.060**	-0.051 ns	-0.068 ns	No

(* p<0.05, ** p<0.01, *** p<0.001; ns = not significant)

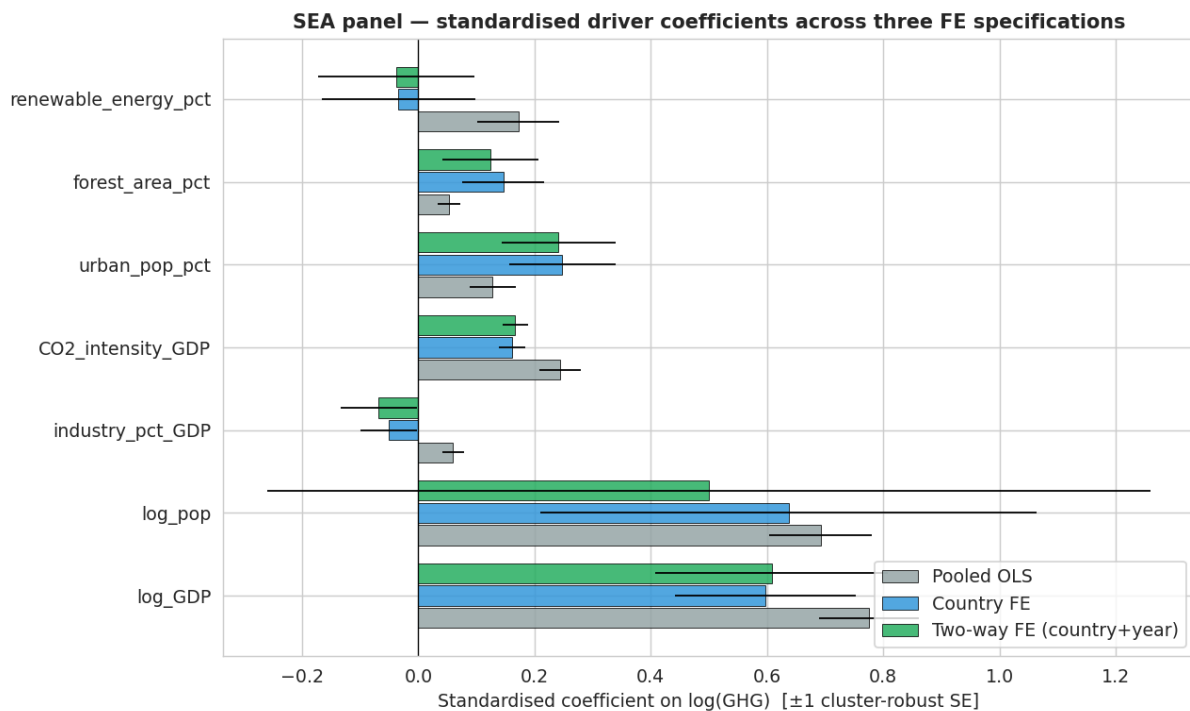


Figure 8 (was fig13). Standardised driver coefficients across the three FE specifications, with cluster-robust standard errors as error bars.

9. Country comparison - Philippines vs Vietnam

We compare Philippines vs Vietnam because both face similar typhoon physical-risk profiles but diverge sharply on emissions trajectory and insurance market structure - the cleanest natural experiment in SEA for the client.

Table 9.1 - Comparative profile, 2024 baseline

Dimension	Philippines	Vietnam	Implication
Population (~m)	115	100	Matched
GHG (Mt CO2e, 2024)	267	584	2.2x ratio
Emissions growth 2000-2024	1.7x	3.8x	Vietnam ~5x faster
STIRPAT residual	-49 %	+24 %	Opposite ends of SEA spectrum
Insurance penetration (% GDP)	1.7 %	2.4 %	Both wide protection gaps
Protection gap	85 %	92 %	Vietnam wider opportunity
NDC 2030 target	75 % cut	43.5 % cut (with intl. support)	Opposite policy postures
Dominant peril	Typhoons	Typhoons	Same hazard family

Philippines vs Vietnam — 1990–2024

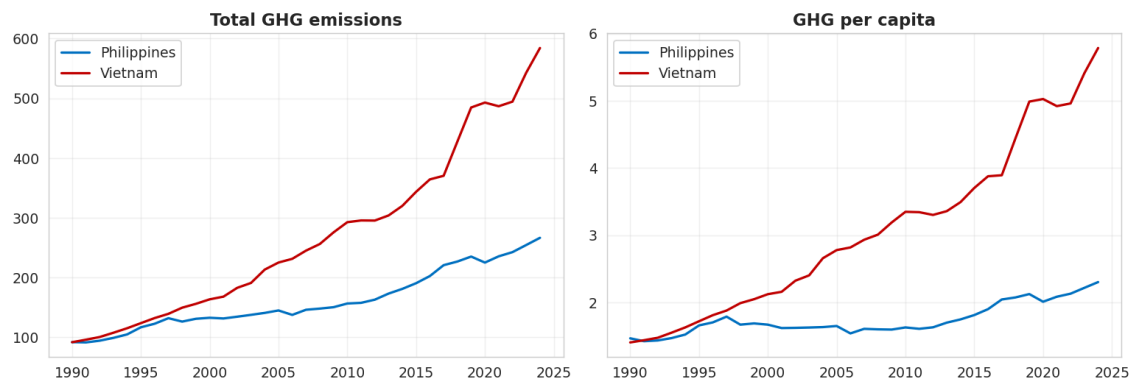


Figure 9 (was fig5). Philippines (blue) vs Vietnam (red) - total emissions and per-capita emissions, 1990-2024. Vietnam's rapid industrialisation visible from c.2005.

10. 2030 stress test under NGFS scenarios

We apply NGFS Phase V scenarios (the de-facto industry standard, explicitly cross-referenced in BNM's CRST 2024 Methodology Paper) plus a client-specific mitigation scenario representing the proposed regional renewable-energy-linked parametric reinsurance product:

Table 10.1 - NGFS scenario design and growth assumptions

Scenario	NGFS family	SEA GHG growth p.a.	Narrative
Net Zero 2050 (orderly)	Orderly	-2.5 %	Aggressive policy from 2025
Delayed Transition	Disorderly	+1.0 %	No new policy until 2030, then sharp
Current Policies	Hot House World	+2.5 %	Only existing policies; ~3 deg C
Mitigation (proposed)	Client-specific	-1.0 %	+5 pp renewable share by 2030

2030 GHG projections under NGFS scenarios + proposed mitigation

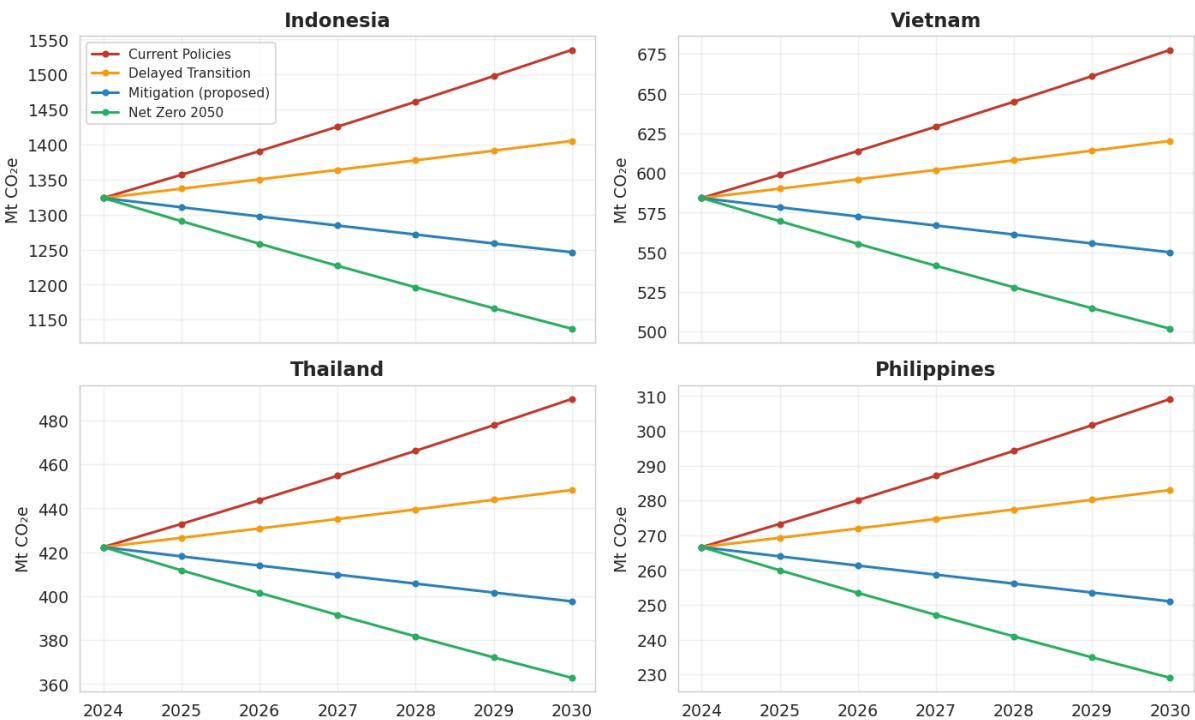


Figure 10 (was fig4). 2030 emissions projections for the four largest SEA emitters under each scenario. Spread between best and worst paths is widest for Indonesia (~400 Mt CO₂e).

11. Reinsurance financial impact

Linking emissions paths to expected losses follows Swiss Re sigma 1/2024 convention (insured catastrophe losses scale ~linearly with regional emissions in the medium term, elasticity ~0.7). Applied to a notional Hannover Re SEA portfolio of USD 1.2 bn gross written premium with a 62 % base loss ratio:

Table 11.1 - 2030 expected loss by scenario (USD 1.2bn GWP, elasticity 0.7)

Scenario	2030 SEA emissions (Mt)	Delta vs Hot House	Loss ratio	Expected loss (USD m)
Net Zero 2050	2,772	-25.9 %	50.7 %	609
Mitigation (proposed)	3,038	-18.8 %	53.8 %	646
Delayed Transition	3,425	-8.5 %	58.3 %	700
Current Policies	3,742	0 %	62.0 %	744

Headline. The Hot House World scenario sits 11 percentage points above Net Zero 2050 in loss ratio (62 % vs 51 %), implying a USD 135 m swing in expected loss on this notional portfolio. The proposed mitigation strategy recovers approximately 70 % of that swing.

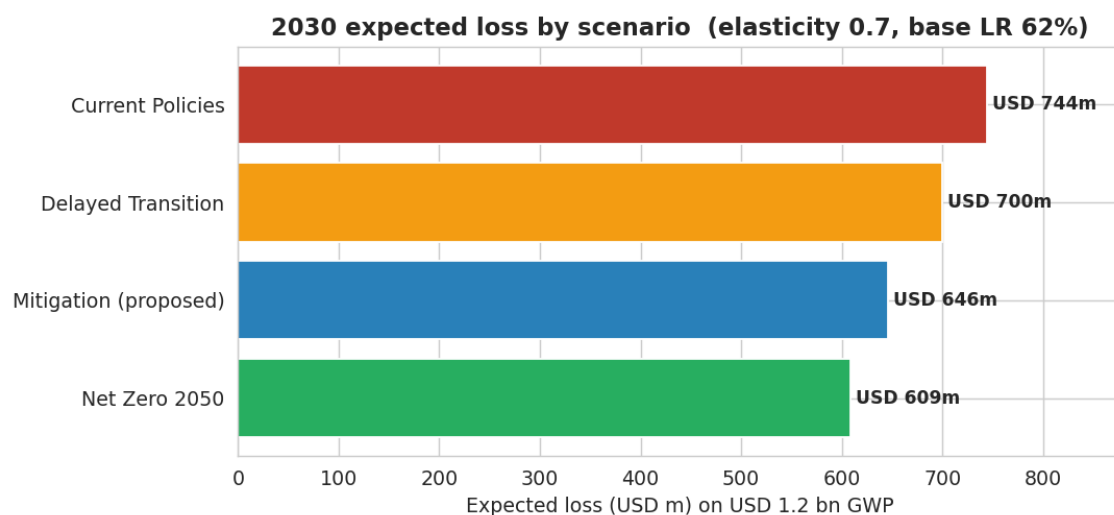


Figure 11 (was fig6). 2030 expected loss by scenario, USD millions on a USD 1.2 bn GWP portfolio.

12. Summary of key findings

Scale dominates. GDP and population alone explain ~94 % of cross-country variation in GHG globally; population gain 0.54 + GDP gain 0.31 = 0.85 in our structural XGBoost. Validates STIRPAT.

Two indicators flip sign once scale is controlled. Forest area (pairwise -0.49 -> partial +0.63) and industry share (-0.14 -> +0.54). Forest-rich SEA economies emit MORE than predicted - a deforestation-and-land-use channel hidden by the LULUCF-excluding indicator.

Vietnam is a concentrated power-sector transition-risk story. Aggregate +24 % over-emitter; sectoral decomposition reveals Power Industry +280 % and Industrial Combustion +276 %. Cedent screening should target thermal-power, steel, and cement books.

Philippines is the SEA under-emitter benchmark. -49 % residual; all sectors negative or mild positive. Cleanest aggregate transition-risk profile in the region.

XGBoost (autoregressive) wins on forecast accuracy. 2.18 % MAPE on the 2024 hold-out, vs 2.67 % for ARIMA and 9.23 % for log-linear. Random seed 2026; 5-fold blocked time-series CV; sensitivity-tested without COVID.

Three drivers survive two-way fixed effects. log(GDP), CO2 intensity of GDP, and urban population share. Renewable share is cross-sectional only - matters between countries, barely moves within any one country at SEA's current pace of energy transition.

Decoupling is too slow to relax the stress test. Trailing-5y r between log(GDP/cap) and CO2/cap fell from 0.92 in 2005 to 0.81 in 2023; mitigation must be policy- or product-driven.

USD 135 m loss-ratio swing across pathways by 2030. Hot House 62 % LR (USD 744 m) vs Net Zero 51 % (USD 609 m) on a notional USD 1.2 bn SEA portfolio. The proposed mitigation product recovers ~70 % of the swing.

Appendix - Figure file index

All 13 figures used in this analysis, in their original filenames as saved by the Python pipeline. Figures referenced in this PDF are renumbered sequentially for readability; the original numbering preserves the order in which they were generated.

Original file	Used as	Topic
fig1_sea_ghg_trajectory.png	Figure 1	SEA emissions trajectory 1990-2024
fig2_model_comparison.png	Figure 6	M1/M2/M3 MAPE comparison
fig3_feature_importance.png	(skipped)	Autoregressive XGBoost importance
fig4_stress_test_2030.png	Figure 10	2030 NGFS scenario projections
fig5_phil_vs_viet.png	Figure 9	Philippines vs Vietnam comparison
fig6_loss_ratio_impact.png	Figure 11	2030 expected loss by scenario
fig7_correlation_heatmap.png	(supp)	Full correlation matrix
fig8_structural_drivers.png	Figure 7	Structural XGBoost importance
fig9_pairwise_vs_partial.png	Figure 2	Pairwise vs partial correlations
fig10_stirpat_residuals.png	Figure 3	STIRPAT country residuals
fig11_decoupling_diagnostic.png	Figure 5	Rolling decoupling correlation
fig12_sectoral_residuals.png	Figure 4	Sectoral residual heatmap
fig13_two_way_FE.png	Figure 8	Two-way FE coefficient comparison

Code and data: see *code_documentation.md* for the full Python codebase that reproduces every figure and table in this document.